

LINUX DAY 2 CLASS WORK

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Day-2: File Ownership and Access Control

Topics: chown, chgrp, ownership + permissions

Part A — Ownership Investigation (10 Minutes)

Create a directory named `ownership_lab`.

Inside it, create:

`incident_report.txt`

`network.log`

`credentials.txt`

Display the detailed information for these files.

For each file, identify:

Owner

Group

Owner permissions

Group permissions

Others permissions

Determine the UID and GID of the current user.

Compare the UID/GID information with the ownership displayed for the newly created files.

Explain the difference between file ownership and file permissions.

```
siva@linux: ~/Desktop/owner_lab
siva@linux:~$ sudo groupadd developers
groupadd: group 'developers' already exists
siva@linux:~$ cat /etc/passwd/group
cat: /etc/passwd/group: Not a directory
siva@linux:~$ cat /etc/group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:syslog
tty:x:5:
disk:x:6:
lp:x:7:
mail:x:8:
news:x:9:
uucp:x:10:
man:x:12:
proxy:x:13:
kmem:x:15:
```

```

siva@linux:~/Desktop/owner_lab$ sudo chown alice incident.txt
[sudo: authenticate] Password:
siva@linux:~/Desktop/owner_lab$ sudo chown bob network.log
siva@linux:~/Desktop/owner_lab$ sudo chown charlie credentials.txt
siva@linux:~/Desktop/owner_lab$ ls -l
total 0
-rw-rw-r-- 1 charlie siva 0 Sep  3 07:04 credentials.txt
-rw-rw-r-- 1 alice   siva 0 Sep  3 07:04 incident.txt
-rw-rw-r-- 1 bob     siva 0 Sep  3 07:04 network.log
siva@linux:~/Desktop/owner_lab$ chown alice network.log
chown: changing ownership of 'network.log': Operation not permitted (os error 1)
siva@linux:~/Desktop/owner_lab$ sudo chown alice:security incident.txt
siva@linux:~/Desktop/owner_lab$ ls -l incident.txt
-rw-rw-r-- 1 alice security 0 Sep  3 07:04 incident.txt
siva@linux:~/Desktop/owner_lab$ sudo chgrp developers network.log
siva@linux:~/Desktop/owner_lab$ sudo chgrp security credentialst.txt
chgrp: cannot dereference 'credentialst.txt': No such file or directory
siva@linux:~/Desktop/owner_lab$ sudo chgrp security credentials.txt
chgrp: cannot dereference 'credentials.txt': No such file or directory
siva@linux:~/Desktop/owner_lab$ sudo chgrp security credentialst.txt
siva@linux:~/Desktop/owner_lab$ mv credentialst.txt credentials.txt
siva@linux:~/Desktop/owner_lab$ ls
credentials.txt incident.txt network.log
siva@linux:~/Desktop/owner_lab$ ls -l network.log credentials.txt
-rw-rw-r-- 1 charlie security 0 Sep  3 07:04 credentials.txt
-rw-rw-r-- 1 bob     developers 0 Sep  3 07:04 network.log
siva@linux:~/Desktop/owner_lab$ group alice
Command 'group' not found, did you mean:
  command 'groups' from deb coreutils-from-gnu (0.0.0-ubuntu25)
  command 'groups' from deb coreutils-from-uutils (0.0.0-ubuntu25)
  command 'groups' from deb coreutils-from-busybox (0.0.0-ubuntu25)
  command 'groups' from deb coreutils-from-toybox (0.0.0-ubuntu25)
Try: sudo apt install <deb name>
siva@linux:~/Desktop/owner_lab$ groups alice
alice : alice
siva@linux:~/Desktop/owner_lab$ groups bob

```

Part B — chown: Changing Ownership (15 Minutes)

Change the owner of incident_report.txt to alice.

Change the owner of network.log to bob.

Change the owner of credentials.txt to charlie.

Verify each change.

Try changing the owner of a file to another user without using sudo. Record the result.

Explain why an ordinary user is generally not permitted to transfer file ownership to another user.

Using a single command, configure:

incident_report.txt

Owner : alice

Group : security

Verify the resulting ownership.

```

siva@linux:~/Desktop/owner_lab$ ls -l
total 0
-rw-rw-r-- 1 siva siva 0 Sep  3 07:04 credentials.txt
-rw-rw-r-- 1 siva siva 0 Sep  3 07:04 incident.txt
-rw-rw-r-- 1 siva siva 0 Sep  3 07:04 network.log
siva@linux:~/Desktop/owner_lab$ id
uid=1000(siva) gid=1000(siva) groups=1000(siva),27(sudo)
siva@linux:~/Desktop/owner_lab$ sudo chown alice incident.txt
[sudo: authenticate] Password:
siva@linux:~/Desktop/owner_lab$ sudo chown bob network.log
siva@linux:~/Desktop/owner_lab$ sudo chown charlie credentials.txt
siva@linux:~/Desktop/owner_lab$ ls -l
total 0
-rw-rw-r-- 1 charlie siva 0 Sep  3 07:04 credentials.txt
-rw-rw-r-- 1 alice siva 0 Sep  3 07:04 incident.txt
-rw-rw-r-- 1 bob siva 0 Sep  3 07:04 network.log

```

Part C — chgrp: Changing Group Ownership (15 Minutes)

Change the group ownership of network.log to developers.

Change the group ownership of credentials.txt to security.

Verify the changes.

Display the group memberships of alice, bob, and charlie.

Log in as alice and create a file named alice_report.txt.

Attempt to change the group of alice_report.txt to security.

Attempt to change its group to developers.

Compare the results.

Explain how a user's group membership affects their ability to change the group ownership of a file they own.

```

siva@linux: ~/Desktop/owner_lab
~/Desktop/owner_lab

siva@linux:~/Desktop/owner_lab$ sudo chgrp developers network.log
siva@linux:~/Desktop/owner_lab$ sudo chgrp security credentialst.txt
chgrp: cannot dereference 'credentialst.txt': No such file or directory
siva@linux:~/Desktop/owner_lab$ sudo chgrp security credentials.txt
chgrp: cannot dereference 'credentials.txt': No such file or directory
siva@linux:~/Desktop/owner_lab$ sudo chgrp security credentials.txt
siva@linux:~/Desktop/owner_lab$ mv credentials.txt credentials.txt
siva@linux:~/Desktop/owner_lab$ ls
credentials.txt incident.txt network.log
siva@linux:~/Desktop/owner_lab$ ls -l network.log credentials.txt
-rw-rw-r-- 1 charlie security 0 Sep  3 07:04 credentials.txt
-rw-rw-r-- 1 bob developers 0 Sep  3 07:04 network.log
siva@linux:~/Desktop/owner_lab$ group alice
Command 'group' not found, did you mean:
  command 'groups' from deb coreutils-from-gnu (0.0.0-ubuntu25)
  command 'groups' from deb coreutils-from-utils (0.0.0-ubuntu25)
  command 'groups' from deb coreutils-from-busybox (0.0.0-ubuntu25)
  command 'groups' from deb coreutils-from-toybox (0.0.0-ubuntu25)
Try: sudo apt install <deb name>
siva@linux:~/Desktop/owner_lab$ groups alice
alice : alice

```

Part D — Ownership + Permission Investigation (20 Minutes)

Configure:

-rw-r----- alice security incident_report.txt

Identify what operations are available to:

alice

charlie

bob

A user who belongs to neither security nor the file owner

charlie belongs to security. Test whether Charlie can:

Read incident_report.txt

Modify incident_report.txt

Delete incident_report.txt

Record the result of each operation.

Change the permissions to:

-rw-rw----

Repeat the test as charlie.

```
charlie
$ cat /path/incident_re.txt
cat: /path/incident_re.txt: No such file or directory
$ cat /path/incident.txt
cat: /path/incident.txt: No such file or directory
$ cat /path/incident.txt
cat: /path/incident.txt: No such file or directory
$ cat incident.txt
$ cat /home/siva/desktop/owner_lab/incident.txt
cat: /home/siva/desktop/owner_lab/incident.txt: Permission denied
$ echo "x" >> /path/incident.txt
sh: 8: cannot create /path/incident.txt: Directory nonexistent
$ rm /path/incident.txt
rm: cannot remove '/path/incident.txt': No such file or directory
$ exit
siva@linux:~/Desktop/owner_lab$ cat /path/incident.txt
cat: /path/incident.txt: No such file or directory
siva@linux:~/Desktop/owner_lab$ ls
credentials.txt  incident.txt  network.log
siva@linux:~/Desktop/owner_lab$ su charlie
Password:
$ cat /path/incident.txt
cat: /path/incident.txt: No such file or directory
$
```

```

siva@linux:~/Desktop/owner_lab$ sudo chmod 660 incident.txt
[sudo: authenticate] Password:
siva@linux:~/Desktop/owner_lab$ echo "x" >> /path/incident.txt
bash: /path/incident.txt: No such file or directory
siva@linux:~/Desktop/owner_lab$ ls
credentials.txt incident.txt network.log
siva@linux:~/Desktop/owner_lab$ cd incident.txt
bash: cd: incident.txt: Not a directory
siva@linux:~/Desktop/owner_lab$

```

```

siva@linux:~/Desktop/owner_lab$ su - bob
Password:
su: warning: cannot change directory to /home/bob: Permission denied
$ cat incident.txt
cat: incident.txt: Permission denied
$

```

Explain why Charlie's access changed even though the owner and group of the file did not change.

Remove bob from the security group.

it checks three separate things independently. who owns the file, what group owns the file, and what permission bits are set for each category (owner / group / others).

1)Ownership answers "whose file is this"

2) permission bits answer "what is each category allowed to do."

why Bob's effective permissions changed.

CMD - sudo gpasswd -d bob security